ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/203738711

Evaluating the Hazardous Failure Rate of majority voting computer architectures by means of Bayesian Network models

Conference Paper · January 2007

CITATIONS
5

READS
304

4 authors:



Francesco Flammini
Mälardalen University
321 PUBLICATIONS **3,530** CITATIONS
SEE PROFILE



Stefano Marrone
Università degli Studi della Campania "Luigi Vanvitelli"
150 PUBLICATIONS **1,460** CITATIONS
SEE PROFILE



Nicola Mazzocca
University of Naples Federico II
328 PUBLICATIONS **3,792** CITATIONS
SEE PROFILE



Valeria Vittorini
University of Naples Federico II
137 PUBLICATIONS **2,133** CITATIONS
SEE PROFILE

All content following this page was uploaded by Francesco Flammini on 02 March 2016.

The user has requested enhancement of the downloaded file.

Software reliability

Risk, Reliability and Societal Safety – Aven & Vinnem (eds)
© 2007 Taylor & Francis Group, London, ISBN 978-0-415-44786-7

Evaluating the Hazardous Failure Rate of majority voting computer architectures by means of Bayesian Network models

F. Flammini
ANSALDO STS – Ansaldo Segnalamento Ferroviario S.p.A., Naples, Italy
Università di Napoli "Federico II" – Dipartimento di Informatica e Sistemistica, Naples, Italy

S. Marrone
ANSALDO STS – Ansaldo Segnalamento Ferroviario S.p.A., Naples, Italy
Seconda Università di Napoli – Dipartimento di Ingegneria dell'Informazione, Aversa (NA), Italy

N. Mazzocca & V. Vittorini
Università di Napoli "Federico II" – Dipartimento di Informatica e Sistemistica, Naples, Italy

ABSTRACT: Safety-critical control systems are usually based on majority voters. In order to assess the compliance of these architectures with international safety standards, the probability of the occurrence of unsafe events should be evaluated by developing and analyzing proper formal models. In this paper we demonstrate that a Bayesian Network (BN) model can be used to evaluate the Mean Time Between Hazardous Events (MTBHE) of voting architectures. The proposed modeling approach is applied to a "2 out of 2" ("2002") voter consisting of independent computing units. The results obtained from the analysis of the BN model of the "2002" voter can be easily extended to evaluate the hazardous failure rate of more complex voting architectures (e.g. Triple Modular Redundant architectures, based on a 2003 voting). Within this context, BNs have several advantages over other traditional approaches (e.g. Petri Nets or Markov Chains): the model can be directly derived from the analysis of the flow-chart describing the dynamic of hazardous failures and its evaluation is much more efficient, as BN solving algorithms are non state-based; moreover, sensitivity analyses can be automatically performed by using the available user friendly BN tools . Finally, the proposed BN model is quite general and can be easily adapted and/or extended to suit specific computing architectures and fault models.

1 INTRODUCTION

Safety-critical control systems are usually based on majority voters. In order to assess the compliance of these architectures with international safety standards, the probability of the occurrence of unsafe events should be evaluated by developing and analyzing proper formal models. Modeling languages for dependability evaluation such as Fault Trees and Stochastic Petri Nets have been widely used for this purpose. In this paper we demonstrate how a Bayesian Network (BN) model can be used to evaluate the Mean Time Between Hazardous Events (MTBHE) of voting architectures. The proposed modeling approach is applied to a "2 out of 2" ("2002") voter consisting of independent computing units. The results obtained from the analysis of the BN model of the "2002" voter can be easily extended to evaluate the hazardous failure rate of more complex voting architectures (e.g. Triple Modular Redundant architectures, based on a 2003 voting). The reference architecture is a safety-critical computer system used for railway control applications. A Generalized Stochastic Petri Net (GSPN) model of the same system is presented in (Amendola 1996) where a safety analysis is performed. The BN model we introduce in this paper has the following advantages over the GSPN approach: (a) the model is more readable and checkable (therefore, less error prone), as it can be directly derived from the analysis of the flow-chart describing the dynamic of Hazardous Failures (HF); (b) the model evaluation is much more efficient, as BN solving algorithms are non state-based; (c) sensitivity analyses can be automatically performed by using the available BN tools (i.e. Most Probable Explanation and Sensitivity to Findings); (d) BN tools are more widespread and user friendly, which is always a point in favor in industrial contexts where the time to learn is a critical factor; (e) the proposed model is more general and can be easily customized and/or extended to

suit specific computing architectures and fault models (for instance, the involved random variables are not required to be exponentially distributed).

The BN model has been validated by using the results obtained from the analysis of the existing GSPN model.

The paper is organized as follows: Section 2 presents a brief summary of related works; Section 3 introduces the reference architecture; Section 4 describes the modeling approach; Section 5 contains some results of the model analysis; Section 5 draws conclusions and provides some hints about future work.

2 RELATED WORKS

An introduction to the dependability of computer systems, including safety related aspects and a reference taxonomy, is provided in (Avizienis 2001). The work presented in this paper is based on the study reported in (Amendola 1996) where the MTBHE of majority voters is evaluated by means of a GSPN model. (Hyunki 2002) also discusses the analysis of a TMR system and in (DeLong 2005) the authors provide a model-based study on hazardous failure of computer architectures using Markov Chains and proper metrics. Dynamic Fault Trees (Dugan 1992) are a flexible formalism for dependability modeling, which is based on combinatorial or Markov methods. Repairable Fault Trees (Codetta Raiteri 2004) can be also used to model behavioral aspects of reliability when the focus is on system's maintenance (Flammini 2005). The application of Bayesian Networks to reliability modeling is introduced in (Portinale 2005). A comparison of the efficiency of different (probabilistic) techniques for safety and dependability assessment can be found in (Bobbio 2001), where it is underlined the trade-off between expressive power and solving efficiency offered by BNs versus Fault Trees and GSPNs. In (Flammini 2006) we applied BNs to the reliability evaluation of a complex railway control system.

3 SYSTEM DESCRIPTION

Several international standards for safety-critical systems (e.g. (CENELEC 1999)) indicate the usage of redundancy and majority voting techniques in implementing safety nucleuses in order to avoid the propagation of hardware failures till the application software layer and the concrete risk of catastrophic system failures. Functional safety is instead qualitatively assessed by means of complementary approaches aimed at defining system functional safety requirements (e.g. Hazard Analysis, see (di Tommaso 2004)).

A schematic representation of a Triple Modular Redundancy (TMR) architecture is shown in Figure 1.

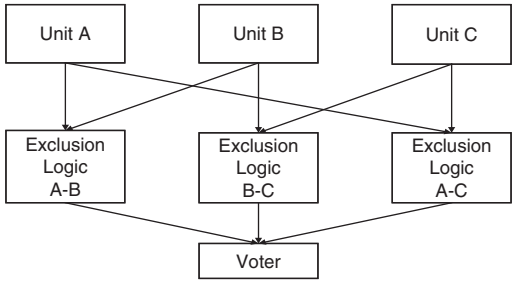


Figure 1. A Triple Modular Redundant architecture.

TMR is a well known technique based on majority voting currently adopted to implement the safety nucleus of most critical control systems. It is based on three independent computing subsystems (or "units") whose output is voted according to a "2 out of 3" ("2003") scheme. Exclusion logic units are external fail-safe devices used to shut-down a failed unit, that is the unit which is not agreeing with the remaining two. When a single unit fails it is excluded and the system continues to operate by relying on two units according to a "2 out of 2" ("2002") voting scheme.

The complete independence of the three units is guaranteed by hardware fault-prevention and fault-tolerance techniques (e.g. galvanic isolation and fiber-optics based communication facilities). At the software level, redundancy codes, design diversity and consistency checks are employed, together with test & diagnostic processes. Thus, the dependability model of these architectures may not take into account common mode of failures. However, the modeling technique presented in this paper also allows to easily evaluate the impact of fault correlation on the hazardous failure rate, whenever required, though this aspect is out of the scope of this work.

4 FAILURE MODELING

The GSPN model for the MTBHE evaluation introduced in (Amendola 1996) has some important limits due to its complexity which have impact on:

- model readability, that is very important for assessment purposes;
- evaluation time, rapidly growing with model size as a state-based solving process is needed;
- maintenance efforts, in terms of skill and time required to modify, update or adapt the model.

We believe that Bayesian Networks can be advantageously employed to address the hazardous failure modeling of "2002" architectures: the resulting model is easier to manage, less error prone when performing maintenance, very quick to evaluate and it also allows for automated sensitivity analyses.

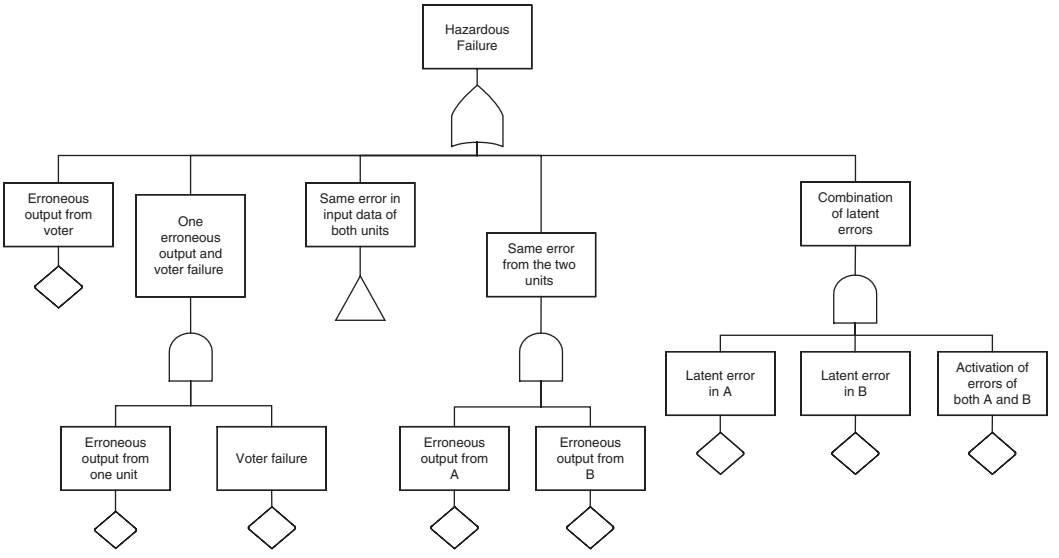


Figure 2. The qualitative Fault Tree for the hazardous failure of the "2002" architecture.

The qualitative Fault Tree depicted in Figure 2 represents five paths of events leading to the occurrence of a hazardous failure, namely:

1. fault in the voter which affects the output;
2. simultaneously occurrence of a wrong output of one faulty unit and a voter failure in detecting unmatching outputs;
3. undetected invalid inputs to both units;
4. undetected invalid outputs from both units;
5. combined activation of two latent errors.

It could be proven (or reasonably justified) that paths 1 and 3 give negligible contributions and thus can be neglected (see (Amendola 1996)).

A more dynamic representation of the behavior of a "2002" system which is a step closer to the real world is provided by the flowchart in Figure 3. It also takes into accounts the dynamic aspects related to the error propagation and the interaction between the two units. While the translation of this flowchart into a GSPN model is not straightforward, its implementation into a Bayesian Network is quite straightforward. The main difference between the two approaches is that the latter is purely probabilistic while the former explicitly considers timing (and therefore state-related) properties. When timing aspects must be introduced in a probabilistic model, one possible way is to make some safe assumptions, i.e. by considering the worst-case conditions in the choice of the net parameters. For instance, if there is a possibility for the on-line diagnostic checks to detect latent errors, the safe assumption consists in neglecting such possibility: if the evaluation result is compliant with system safety requirements, then we

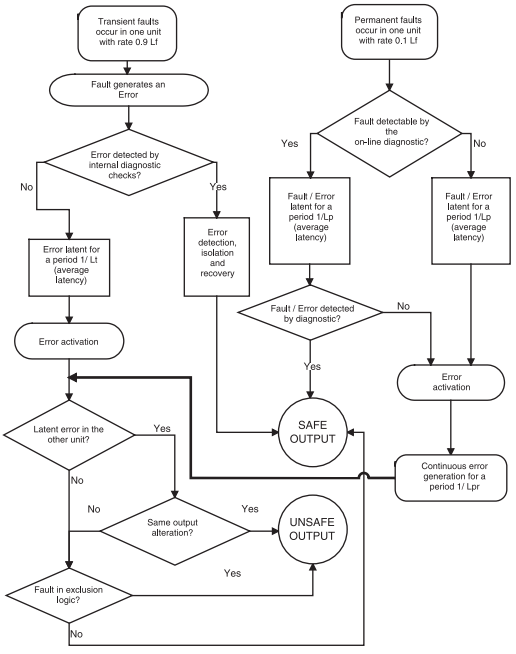


Figure 3. The hazardous failure flow-chart.

can be satisfied with such result; otherwise, we need to refine the model. In the example, the refinement can be performed by means of an "off-line" analysis and prediction of the detection probability of the diagnostic checks during the reference time interval, which is 1 hour in case of the MTBHE model; the result of the

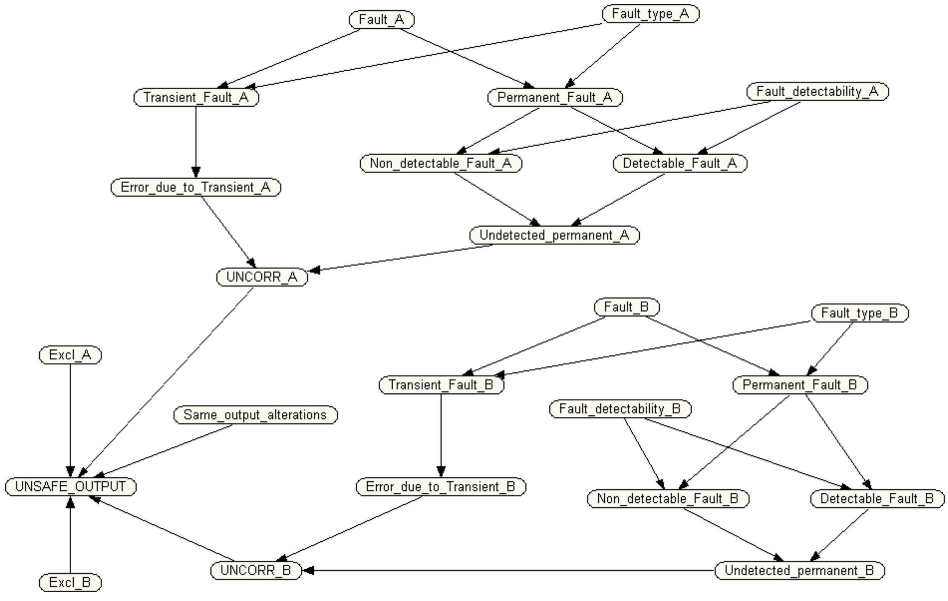


Figure 4. The Bayesian Network model of the "2002" computer architecture.

Table 1. The Conditional Probability Table for the UNSAFE OUTPUT event.

True	False	Same Output Alt.	UNCORR_A	UNCORR_B	Excl_A	Excl_B
1	0	True	True	True	True	True
					True	False
					False	True
					False	False
				False	True	True
					True	False
0	1				False	True
0	1				False	False
1	0		False	True	True	True
0	1				True	False
1	0				False	True
					False	False
0	1			False	True	True
					True	False
					False	True
					False	False
1	0	False	True	True	True	True
1	0				True	False
1	0				False	True
0	1				False	False
1	0			False	True	True
1	0				True	False
0	1				False	True
0	1				False	False
1	0		False	True	True	True
0	1				True	False
1	0				False	True
0	1				False	False
0	1			False	True	True
					True	False
					False	True
					False	False

Table 2. Bayesian Network parameters and results.

Stochastic Var.	Meaning	Value	Probability
Detectable_Fault_A	A detectable fault occurring in Unit A	True	1.4999e-006
Detectable_Fault_B	A detectable fault occurring in Unit B	True	1.4999e-006
Error_due_to_Transient_A	An error due to a transient fault occurring in Unit A	True	1.4995e-006
Error_due_to_Transient_B	An error due to a transient fault occurring in Unit B	True	1.4995e-006
Excl_A	A failure occurring in the Exclusion Logic of Unit A	True	1e-010
Excl_B	A failure occurring in the Exclusion Logic of Unit B	True	1e-010
Fault_A	A fault occurring in Unit A	True	1.6666e-005
Fault_B	A fault occurring in Unit B	True	1.6666e-005
Fault_detectability_A	The possibility of detecting a fault occurring in Unit A	Detectable	0.9
Fault_detectability_B	The possibility of detecting a fault occurring in Unit B	Detectable	0.9
Fault_type_A	The fault type of Unit A (Transient or Permanent)	Transient	0.9
Fault_type_B	The fault type of Unit B (Transient or Permanent)	Transient	0.9
Non_detectable_Fault_A	A non detectable fault occurring in Unit A	True	1.6666e-007
Non_detectable_Fault_B	A non detectable fault occurring in Unit B	True	1.6666e-007
Permanent_Fault_A	A permanent fault occurring in Unit A	True	1.6666e-006
Permanent_Fault_B	A permanent fault occurring in Unit B	True	1.6666e-006
Same_output_alterations	The two units produce the same modification of their output	True	0.1
Transient_Fault_A	A transient fault occurring in Unit A	True	1.4999e-005
Transient_Fault_B	A transient fault occurring in Unit B	True	1.4999e-005
UNCORR_A	An uncorrect output occurring in Unit A	True	2.1912e-006
UNCORR_B	An uncorrect output occurring in Unit B	True	2.1912e-006
Undetected_permanent_A	An undetected permanent fault occurring in Unit A	True	6.9164e-007
Undetected_permanent_B	An undetected permanent fault occurring in Unit B	True	6.9164e-007
UNSAFE_OUTPUT	An unsafe output occurring in the system	True	4.8056e-013

analysis is finally embedded in the model as a plain probability.

The Bayesian Network model of a "2002" computer architecture is depicted in Figure 4. It is a quite self-explaining model which can be easily derived from the flowchart of Figure 3.

The Conditional Probability Table (CPT) for the random variable modeling the occurrence of the UNSAFE FAILURE event is shown in Table 1.

The parameters and the probabilities used to populate the model have been listed in Table 2. The (reported) values are fixed in this study, since they are related to hardware fault probabilities, design choices and measured data.

The "2003" failure model can be derived from the "2002" model:

$$\mathrm{MTBHE}_{2003} = 1/3 \cdot \mathrm{MTBHE}_{2002}$$

In fact, the behavior of a "2003" system can be described by three "2002" systems: units A AND B, units B AND C, units A AND C.

5 MODEL EVALUATION

The model must be evaluated in order to demonstrate the compliance to system safety requirements. For instance, (Unisig 2005) reports the safety requirements of critical railway control systems (i.e. ERTMS/ETCS, see (Unisig 2002)). In this document, system structural safety requirement of $2 \cdot 10^{-9}$ hazardous failures per hour (i.e. THR, Tolerable Hazard Rate) is apportioned on the ground subsystem for one half (i.e. 10^{-9} HF/h), the MTBHE being obviously the inverse of the THR. Generally speaking, values in the order of 10^{-9} are common when defining quantitative safety requirements. Quite intuitively, keeping all subsystems several orders of magnitudes below this threshold gives good chances to respect the overall safety goal.

Coming back to the model, please note that the probability of a permanent detectable error activation before the diagnostic is executed, considering the reference time interval of 1 h and the diagnostic cycle time, can be obtained as follows:

$$0.9 \cdot (1 - 1000/3600) = 0.9 \cdot (1 - 0.27777) =$$

$$= 0.9 \cdot 0.72222 = 0.65$$

The tool we used for model construction and evaluation is Netica (Netica 2006). The results of the analysis are reported in Table 2, where the last row (shaded in grey) highlights the main result, consisting in the evaluated Hazard Rate (HR) of the system.

A sensitivity analysis has also been performed on the model in order to validate it against the results presented in (Amendola 1996). For the sake of brevity, we omit here the results of such analyses.

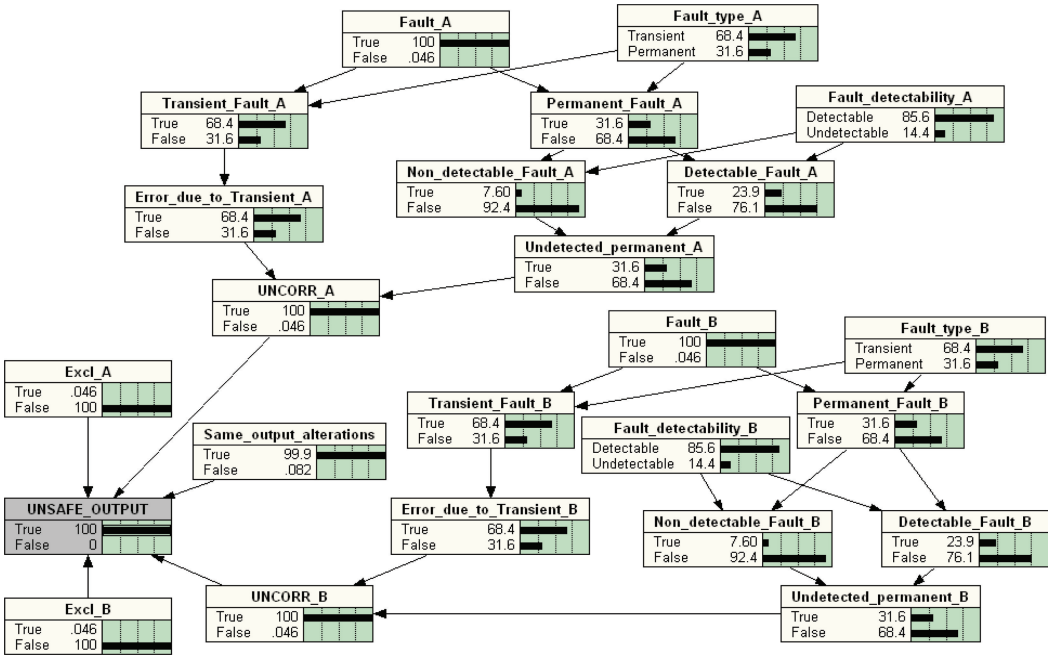


Figure 5. Model evaluation by means of "a posteriori" probabilities.

Quite obviously:

$$MTBHE_{2003} = 1 / (3 \cdot HR_{2002}) = 6.9362 \cdot 10^{11} \text{ h}$$

which is largely compliant to system safety requirements, also considered the other possible causes of hazardous failures.

Compared with the data provided in (Amendola 1996), these results have served to validate our model.

Finally, Figure 5 reports the results of an "a posteriori" probability evaluation (with the UNSAFE_OUTPUT event value fixed to TRUE), highlighting the power and flexibility of BN and of the related analysis tools.

6 CONCLUSIONS

In this paper we have shown how Bayesian Network models can be employed in order to model a quite general case-study, consisting in a Triple Modular Redundant computing subsystem used in safety-critical control applications. The results we have obtained support the use of BNs for an easy and effective representation of the probabilistic hazardous failure model, as they feature several advantages with respect to other formalisms.

In the future, we intend to employ Repairable Fault Trees and/or Dynamic Bayesian Networks (allowing to explicitly model state-base aspects, see (Montani 2005)) in order to extend the model for the evaluation of imperfect maintenance: after a system shut-down due to a permanent hardware fault, system is possibly restarted without substituting the faulty section (faults could be possibly non-diagnosable and happening during system down-time). Stochastic Well-formed nets could also be employed in order to reduce GSPN model complexity by exploiting symmetries using model folding (Chiola 1993).

REFERENCES

Ajmone Marsan, M.; Balbo, G.; Conte, G.; Donatelli, S. & G. Franceschinis 1995. *Modeling with Generalized Stochastic Petri Nets*: J. Wiley.

Amendola, A.M.; Impagliazzo, L.; Marmo, P.; Mongardi, G. & Sartore, G. 1996. Architecture and Safety Requirements of the ACC Railway Interlocking System. *IEEE Proc. 2nd Annual Int. Computer Performance & Dependability Symposium* (IPDS'96), Urbana Champaign, IL, USA: 21–29.

Avizienis, A.; Laprie, J.C. & B. Randel 2001. *Fundamental Concepts of Dependability*, LAAS Report no. 01–145.

Bobbio, A.; Bologna, S.; Ciancamerla, E.; Franceschinis, G.; Gaeta, R.; Minichino, M. & Portinale, L. 2001. Comparison of Methodologies for the Safety and Dependability Assessment of an Industrial Programmable Logic Controller. *Proceedings of the European Safety & Reliability Conference* (ESREL 2001), Turin, Italy: 411–418.

CENELEC 1999. *EN 50126 Railways Applications – The specification and demonstration of Reliability, Maintainability and Safety (RAMS)*.

Chiola, G.; Dutheillet, C.; Franceschinis, G. & Haddad, S. 1993. Stochastic Well-Formed Colored Nets and Symmetric Modeling Applications. *IEEE Transactions on Computers*, vol. 42: 1343–1360.

Codetta Raiteri, D.; Iacono, M.; Franceschinis, G. & Vittorini, V. 2004. Repairable Fault Tree for the Automatic Evaluation of Repair Policies. *Proceedings of the Int. Conf. on Dependable Systems and Networks 2004 (DSN 2004)*: 659–668.

DeLong, T.A.; Smith, D.T. & Johnson, B.W. 2005. Dependability metrics to assess safety-critical systems. *IEEE Transactions on Reliability*, vol.54, no. 3: 498–505.

di Tommaso, P.; Esposito, R.; Marmo, P. & Orazzo, A. 2003. Hazard Analysis of Complex Distributed Railway Systems. *Proceedings of 22nd International Symposium on Reliable Distributed Systems*, Florence: 283–292.

Dugan, J.B.; Bavoso, S.J. & Boyd, M.A. 1992. Dynamic Fault-Tree Models for Fault Tolerant Computer Systems. *IEEE Transactions on Reliability*, vol. 41, 1992: 363–377.

Flammini, F.; Iacono, M.; Marrone, S. & Mazzocca, N. 2005. Using Repairable Fault Trees for the evaluation of design choices for critical repairable systems. *Proceedings of the 9th IEEE International Symposium on High Assurance Systems Engineering* (HASE2005), Heidelberg, Germany, October 12–14: 163–172.

Flammini, F.; Marrone, S.; Mazzocca, N. & Vittorini, V. 2006. Modelling System Reliability Aspects of ERTMS/ETCS by Fault Trees and Bayesian Networks. *Safety and Reliability for Managing Risk: Proceedings of the 15th European Safety and Reliability Conference* (ESREL 2006), Estoril, Portugal: 2675–2683.

Hyunki, K.; Hyung-Joon, J.; Keyseo, L. & Hyuntae, L. 2002. The design and evaluation of all voting triple modular redundancy system. *Proceedings Annual Reliability and Maintainability Symposium*: 439–444.

Montani, S.; Portinale, L. & Bobbio, A. 2005. Dynamic Bayesian Networks for Modeling Advanced Fault Tree Features in Dependability Analysis. *Proc. of European Safety and Reliability Conference* (ESREL 2005), Tri City, Poland: 1415–1422.

NASA Office of Safety and Mission Assurance 2002. *Fault Tree Handbook with Aerospace Applications*, ver. 1.1.

Netica web site 2006: http://www.norsys.com/netica.html.

Portinale, L.; Bobbio, A. & Montani, S. 2005. From AI to Dependability: Using Bayesian Networks for Reliability Modeling and Analysis. *Modern Statistical and Mathematical Methods in Reliability*, Series on Quality, Reliability and Engineering Statistics, vol. 10: 365–382.

UNISIG 2002. *ERTMS/ETCS – Class 1 SRS Issue 2.2.2, Subset-026.*

UNISIG 2005. *ERTMS/ETCS – Class 1 Safety Requirements Issue 2.2.11, Subset-091.*